

The American Journal of Sports Medicine

<http://ajs.sagepub.com/>

Effect of Matching or Overconstraining Knee Laxity During Anterior Cruciate Ligament Reconstruction on Knee Osteoarthritis and Clinical Outcomes: A Randomized Controlled Trial With 84-Month Follow-up

Matthew R. Akelman, Paul D. Fadale, Michael J. Hulstyn, Robert M. Shalvoy, Arlene Garcia, Kaitlyn E. Chin, Jeffrey Duryea, Gary J. Badger, Glenn A. Tung and Braden C. Fleming

Am J Sports Med published online April 19, 2016

DOI: 10.1177/0363546516638387

The online version of this article can be found at:

<http://ajs.sagepub.com/content/early/2016/04/19/0363546516638387>

Published by:



<http://www.sagepublications.com>

On behalf of:

[American Orthopaedic Society for Sports Medicine](#)



Additional services and information for *The American Journal of Sports Medicine* can be found at:

Published online April 19, 2016 in advance of the print journal.

P<P

Email Alerts: <http://ajs.sagepub.com/cgi/alerts>

Subscriptions: <http://ajs.sagepub.com/subscriptions>

Reprints: <http://www.sagepub.com/journalsReprints.nav>

Permissions: <http://www.sagepub.com/journalsPermissions.nav>

>> [OnlineFirst Version of Record](#) - Apr 19, 2016

[What is This?](#)

Effect of Matching or Overconstraining Knee Laxity During Anterior Cruciate Ligament Reconstruction on Knee Osteoarthritis and Clinical Outcomes

A Randomized Controlled Trial With 84-Month Follow-up

Matthew R. Akelman,^{*} BA, Paul D. Fadale,[†] MD, Michael J. Hulstyn,[†] MD, Robert M. Shalvoy,[‡] MD, Arlene Garcia,[†] BS, Kaitlyn E. Chin,[†] BS, Jeffrey Duryea,[§] PhD, Gary J. Badger,^{||} MS, Glenn A. Tung,[¶] MD, and Braden C. Fleming,^{†#} PhD
Investigation performed at Rhode Island Hospital, Brown University, Providence, Rhode Island, USA

Background: The “initial graft tension” applied at the time of graft fixation during anterior cruciate ligament (ACL) reconstruction surgery modulates joint contact mechanics, which in turn may promote posttraumatic osteoarthritis (OA).

Purpose/Hypotheses: The study objectives were to compare clinical, functional, patient-reported, and OA imaging outcomes between 2 different initial laxity-based graft tension cohorts and a matched uninjured control group as well as to evaluate the effects of laxity-based graft tension on OA development at 84-month follow-up. The 2 laxity-based tension protocols were (1) to restore normal anteroposterior (AP) laxity at the time of surgery relative to the contralateral uninjured knee (low-tension group) or (2) to overconstrain AP laxity by 2 mm relative to the contralateral uninjured knee (high-tension group). The hypotheses were that (1) the high-tension group would have improved outcomes and decreased OA compared with the low-tension group after 84 months, and (2) the outcomes for the high-tension group would be equivalent to those for an age-, sex-, race-, and activity-matched group of control participants with uninjured knees.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: Patients had their ACLs reconstructed with either a bone–patellar tendon–bone or 4-stranded hamstring autograft, and outcomes were compared with a matched control group. Outcomes were evaluated preoperatively and at 60 and 84 months postoperatively and included clinical (KT-1000 arthrometer AP laxity measurement and International Knee Documentation Committee [IKDC] examination score), functional (1-legged hop for distance and knee extensor torque), patient-reported (Knee injury and Osteoarthritis Outcome Score [KOOS], Short Form–36 [SF-36], and patient satisfaction survey), and OA imaging (measurement of joint space width [JSW], Osteoarthritis Research Society International [OARSI] radiographic score, and Whole-Organ Magnetic Resonance Imaging Score [WORMS]) components. Repeated-measures analyses of variance were used to evaluate differences in outcomes between the treatment groups and the control group.

Results: There were significant differences between the 2 tension groups in 1 of 5 KOOS subscales (sports and recreation; $P = .04$) and 2 of 8 SF-36 subscales (vitality, mental health; $P < .04$) at 84 and 60 months, respectively. Both tension groups scored significantly worse than the control group in the IKDC examination ($P < .001$), 1-legged hop ($P \leq .017$), KOOS quality of life and symptoms subscales ($P < .03$), and OARSI radiographic score ($P \leq .02$) at 84 months. The low-tension group performed significantly worse than the control group on the KOOS pain subscale ($P = .03$), SF-36 general health and social functioning ($P < .04$), OARSI radiographic score ($P < .001$), and WORMS ($P = .001$), while the high-tension group had statistically different results than the control group in AP knee laxity ($P < .001$), radiographic JSW ($P = .003$), and OARSI radiographic score ($P = .02$) as well as significantly more subsequent knee injuries ($P = .02$) at 84 months.

Conclusion: The results do not support the hypotheses that the high-tension group would have improved outcomes when compared with the low-tension group after 84 months of healing or that the outcomes for the high-tension group would be equivalent

to those for the matched control group. While there were minor differences in patient-reported outcomes between the 2 laxity-based tension groups, all other outcomes were similar.

Registration: NCT00434837

Keywords: anterior cruciate ligament (ACL); autograft; reconstruction; tension; outcomes

Anterior cruciate ligament (ACL) reconstruction is commonly performed to restore joint function. However, evidence suggests that it may not reduce the risk of posttraumatic osteoarthritis (OA).¹ The “initial graft tension” applied at the time of graft fixation modulates joint contact mechanics, which in turn may promote knee OA. We previously published a study looking at the short-term (eg, 36-month) outcomes of 2 different initial graft tension cohorts, using a “laxity-based” approach to set the initial graft tension.²⁰ There were no detectable differences in clinical, functional, patient-reported, and OA imaging outcomes between the 2 laxity-based graft tension groups, and neither group returned to the levels of the matched control group. However, it remained unknown whether these outcomes would continue to improve with time and become equivalent to those of the control group with long-term healing. Considering that the 2 laxity-based initial graft tension protocols produced different initial joint contact conditions,¹⁹ that autografts continue to remodel after 3 years of healing,⁷ and that sufficient time is required before OA can be detected,²⁴ long-term follow-up of these cohorts was needed.

The objectives of this prospective randomized controlled trial (NCT00434837) were to compare clinical, functional, patient-reported, and OA imaging outcomes between 2 different initial laxity-based graft tension cohorts and a matched uninjured control group and to evaluate the effects of laxity-based graft tension on OA development at 84-month (7-year) follow-up. The matched control group was included to provide baseline data to compare OA progression and other outcome measures. The 2 laxity-based tension protocols under investigation were (1) to set the initial graft tension to restore normal anteroposterior (AP) laxity at the time of surgery relative to the contralateral uninjured knee (low-tension group) or (2) to set the initial graft tension to constrain AP laxity by 2 mm relative to the contralateral knee (high-tension group). The clinical outcomes included the KT-1000 arthrometer measurement,¹² International Knee Documentation Committee (IKDC) examination score,²⁵ and the

number of subsequent knee injuries. The functional outcomes included the 1-legged hop for distance³⁷ and knee extensor torque.³³ The patient-reported outcomes included the Knee injury and Osteoarthritis Outcome Score (KOOS),³⁹ Short Form-36 (SF-36 [v2]),⁴³ and a patient satisfaction survey. The OA imaging outcomes included radiographic measures of joint space width (JSW),²⁶ Osteoarthritis Research society International (OARSI) radiographic score,² and Whole-Organ Magnetic Resonance Imaging Score (WORMS).³⁴ We hypothesized that (1) the high-tension group would have improved clinical, functional, patient-reported, and OA imaging outcomes compared with the low-tension group after 84 months and (2) the outcomes for the high-tension group would be equivalent to those for a sex-, race-, and age-matched control group.

METHODS

Trial Design

This prospective randomized controlled trial is an extension of our short-term analysis²⁰ and includes follow-up data obtained at 60 and 84 months. The institutional review boards of Rhode Island Hospital, Miriam Hospital, Memorial Hospital of Rhode Island, and Brown University approved the study, and all participants granted informed consent. The study was designed as an intent-to-treat analysis. A statistician performed the randomization. Although this study was designed as a double-blind trial, the operating surgeon was informed of the randomization assignment at the time of graft implantation. Although blinding was unveiled by the statistician to analyze the 36-month data, investigators were blinded to the participant group assignment for all 60- and 84-month follow-up assessments. An a priori power analysis determined that 45 participants per group would be necessary to detect a mean difference in the JSW of 5% ($1 - \beta = .80$; $\alpha = .05$).²⁰ Based on this sample size, the study was powered to detect a 16% difference in AP laxity, 12%

#Address correspondence to Braden C. Fleming, PhD, Brown University, 1 Hoppin Street, Coro West, Suite 404, Providence, RI 02903, USA (email: Braden_Fleming@brown.edu).

*College of Medicine, Medical University of South Carolina, Charleston, South Carolina, USA.

†Department of Orthopaedics, Rhode Island Hospital, Brown University, Providence, Rhode Island, USA.

‡Department of Orthopaedics, Memorial Hospital of Rhode Island, Brown University, Pawtucket, Rhode Island, USA.

§Department of Radiology, Brigham and Women's Hospital, Harvard University, Boston, Massachusetts, USA.

||Department of Medical Biostatistics, University of Vermont, Burlington, Vermont, USA.

*Department of Diagnostic Imaging, Rhode Island Hospital, Brown University, Providence, Rhode Island, USA.

The contents of this article are solely the responsibility of the authors and do not necessarily represent the official views of the National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS) of the National Institutes of Health (NIH).

One or more of the authors has declared the following potential conflict of interest or source of funding: This project was funded by the NIH (2R01-AR047910 from the NIAMS) and the Lucy Lippitt Endowed Professorship. B.C.F. receives a stipend for serving as an associate editor of *The American Journal of Sports Medicine*.

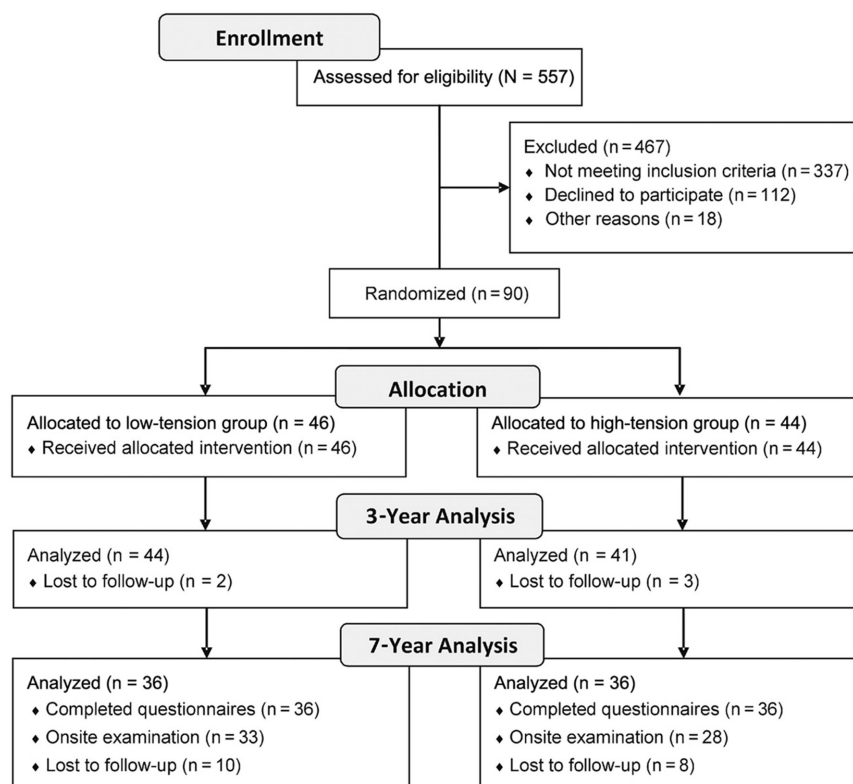


Figure 1. CONSORT diagram of the study design, patient allocation, and loss to follow-up through 7 years.²⁰

difference in the IKDC score, 12% difference in peak isokinetic torque, and 10% difference in the KOOS value.

Participants and Entry Criteria

All patients who presented with an isolated unilateral ACL injury in the clinics of 3 surgeons (P.D.F., M.J.H., R.M.S.) over 3 years (February 2004 to February 2007) were assessed for eligibility as previously described.²⁰ In short, male and female patients aged between 15 and 50 years with a unilateral ACL injury who were candidates for surgical reconstruction with a bone–patellar tendon–bone or 4-stranded hamstring tendon autograft were included. Patients with an ACL injury more than 12 months old, a significant concomitant injury to the menisci or other ligaments, a previous knee injury, or evidence of OA were excluded. Of the 557 patients screened, 355 were excluded for not meeting the inclusion criteria, and 112 declined to participate.²⁰ Of the 90 patients randomized between the 2 tension groups (low-tension group, $n = 46$; high-tension group, $n = 44$), 18 (20%) were lost to follow-up at 84 months (Figure 1). Of the remaining 72 patients, 100% filled out the questionnaires, and 86% returned for the on-site follow-up examination. An uninjured control group, comprising 60 participants who were matched by age, sex, race, and activity level, was also recruited from the local area via advertisement.²⁰ Control participants were excluded if they had a previous knee injury, increased knee laxity, or evidence of OA.²⁰ Twenty-five (42%) of the

control participants were lost to follow-up at 84 months, 32 returned for their 84-month examination, and 3 sent in their questionnaires.

ACL Reconstruction/Initial Graft Tension Protocols

ACLs were reconstructed either with a bone–patellar tendon–bone autograft obtained from the central third of the ipsilateral patellar tendon or a 4-stranded semitendinosus and gracilis autograft.²⁰ Patients selected the graft type, and all surgeons followed the same operative procedures. Grafts were preconditioned with 20 manual tension cycles before fixation. For the patellar tendon grafts, the bone blocks were secured using interference screws. The hamstring tendon grafts were fixed with cortical fixation on the femur and a biodegradable interference screw on the tibia, backed up at the surgeon's discretion with a screw and spiked soft tissue washer. For those patients receiving the low-tension assignment, the grafts were tensioned by firmly pulling on the distal graft with the knee at 0° of knee flexion.^{19,20} For those receiving the high-tension assignment, the grafts were firmly tensioned with the knee at 30° of knee flexion.^{19,20} Tibial fixation was partially engaged for both graft types, and AP laxity at 20° of flexion was checked using the KT-1000S arthrometer (MEDmetric Corp) and compared with that of the contralateral knee under anesthesia. If the targeted laxity value was not achieved within 1 mm, fixation was released, and the tensioning procedure was repeated. The laxity value was rechecked once the

fixation procedure was completed, and this value was used for all subsequent analyses. Postoperatively, all patients followed a standardized rehabilitation program designed to have them return to sport at 6 months.⁶

Clinical Outcomes

AP laxity values for both knees were measured using the KT-1000 knee arthrometer (MEDmetric Corp) preoperatively and at all postoperative time points.¹² Trained sports physical therapists dedicated to the project performed the KT-1000/KT-1000S measurements. Three manual maximum tests were performed and the displacement readings averaged. The difference between legs was compared and reported.

Clinical outcomes were assessed using the 2000 IKDC examination score (<http://www.sportsmed.org>).²⁵ The IKDC evaluates 4 categories: (1) function, (2) symptoms, (3) range of knee motion, and (4) clinical examination; it also rates knees as normal (A), nearly normal (B), abnormal (C), or severely abnormal (D), with the final IKDC rating based on the score of the worst category. Subsequent knee injuries to the ipsilateral or contralateral knee were determined by patient questioning and a medical record review at each follow-up visit. Trained sports physical therapists administered all clinical examinations.

Functional Outcomes

Patients performed the 1-legged hop test for distance 3 times, and the trials were averaged.³⁷ The mean hop distance of the injured leg was normalized to that of the uninjured contralateral leg. Isokinetic strength testing (Biodex 2; Biodex Medical Systems Inc) was performed at 60 deg/s to assess the strength of the extensor muscles of each knee.³³ Peak torques for 5 repetitions were averaged and normalized with respect to body weight and height.

Patient-Reported Outcomes

The KOOS³⁹ and SF-36⁴³ were implemented to assess patient-reported outcomes. The KOOS evaluates 5 domains: (1) quality of life (KOOS-QOL), (2) sports and recreation (KOOS-sport), (3) activities of daily living (KOOS-ADL), (4) symptoms (KOOS-symptoms), and (5) pain (KOOS-pain). For this study, the composite KOOS model developed by Englund et al¹⁸ was additionally used to identify patients with joint arthrosis. Patients with a KOOS-QOL value of ≤ 87.5 and who did not meet the thresholds of at least 2 of the other subscales (ie, KOOS-pain ≤ 86.1 , KOOS-symptoms ≤ 85.7 , KOOS-ADL ≤ 86.8 , and KOOS-sport ≤ 85.0) were designated as having OA using this model.^{18,44} The SF-36 evaluates health status related to physical functioning, role limitations, bodily pain, vitality, social functioning, emotional role, mental health, and general health.⁴³

Activity levels were monitored at the time of recruitment and at each follow-up using the Tegner activity scale.⁴⁰ A supplemental survey, in which participants were asked if they had any subsequent knee injuries that required a doctor's visit, how satisfied they were with their baseline surgical outcome (1 = not satisfied, 10 = very satisfied), and if they

would elect to undergo ACL reconstruction surgery again, was mailed after they completed their 84-month follow-up.

OA Imaging Outcomes

Minimum medial JSW measurements were obtained from standardized radiographs preoperatively and postoperatively using a posteroanterior view of the semiflexed knee.^{8,26} The knee flexion angle was standardized across patients by placing the metatarsal phalangeal joint of the foot in the same plane as the image intensifier, and the knee was flexed forward until the patella touched the image intensifier as previously described.^{8,20} The minimum JSW of the medial compartment was calculated using a validated automated algorithm,^{13,14} which was recently used to measure JSW changes in the Multicenter Orthopaedic Outcomes Network (MOON) "nested" cohort after ACL reconstruction^{26,27} and has been recently shown to be more reliable than the method originally used in the 36-month follow-up of the current study.²⁰

The overall condition of the joint was assessed preoperatively and postoperatively using the modified OARSI radiographic score.²² On the basis of posteroanterior and lateral radiographs of both knees, a grade of 0 (normal) to 4 (severe) was assigned to 2 radiographic features: osteophyte formation and joint space narrowing. In addition, sclerosis, attrition, and ligament calcification were assessed on a dichotomous scale. An experienced musculoskeletal radiologist (G.A.T.) scored all radiographs while blinded to the treatment group.

OA was also assessed using the WORMS.³⁵ The score utilizes magnetic resonance sequences to grade 14 independent features, including cartilage signal and morphology, subarticular bone marrow abnormalities, subarticular cysts, subarticular bone attrition, and marginal osteophytes evaluated in 15 regions. The condition of the menisci, cruciate and collateral ligaments, synovitis, loose bodies, and periarticular cysts were also included. The WORMS results were assessed by an experienced musculoskeletal radiologist (G.A.T.) who was blinded to treatment.

Statistical Analysis

Repeated-measures analyses of variance were used to evaluate differences among treatment groups with respect to temporal changes in outcome variables. The statistical models included fixed factors representing treatment group (low-tension, high-tension, and control) and time. For each control participant, a knee was randomly selected to represent the reconstructed knee for the outcome measures reported as differences or ratios between limbs. If significant interactions were detected, F tests corresponding to simple effects were examined. Pairwise comparisons were performed using the Fisher least significant difference test, and chi-square tests were utilized to compare the frequency of subsequent injuries and the distributions of IKDC scores between groups. A chi-square test was also used to compare the percentage of patients within each group in which the composite KOOS model^{18,44} indicated the presence of OA. All means represent

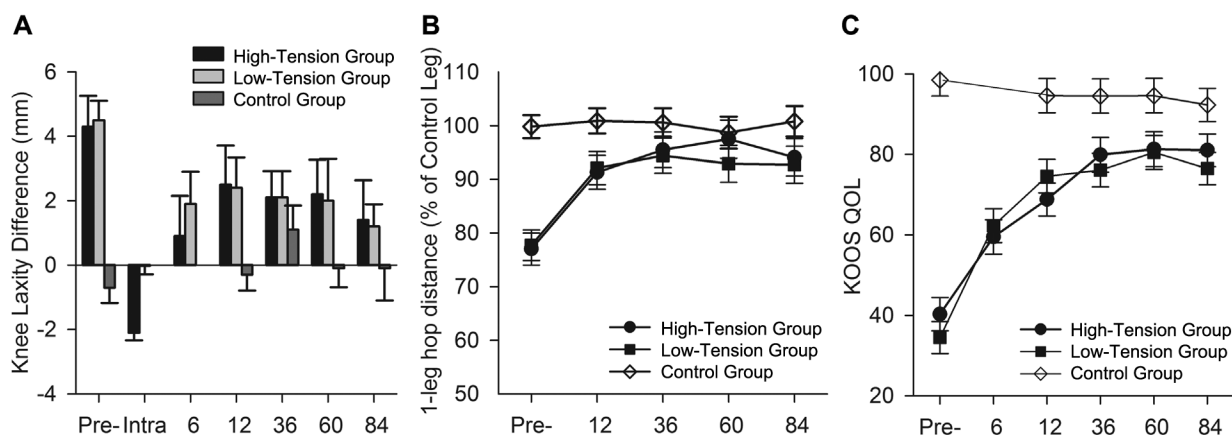


Figure 2. Examples of (A) clinical (KT-1000 arthrometer), (B) functional (1-legged hop for distance), and (C) patient-oriented (Knee injury and Osteoarthritis Outcome Score–quality of life [KOOS-QOL] subscale) outcomes over 84 months.

the least-squares means (95% CIs). $P < .05$ was considered statistically significant.

RESULTS

Patient Characteristics and Intraoperative Findings

At the 84-month visit, 10 patients in the low-tension group, 8 in the high-tension group, and 25 in the control group were lost to follow-up. Patient sex (60% women vs 48% women), age (23 vs 24 years), weight (69 vs 73 kg), time between injury and index surgery (105 vs 114 days), percentage of patients receiving patellar tendon grafts (61% vs 67%), and percentage of patients with minor meniscal injuries (41% vs 41%) or minor chondral lesions (23% vs 17%) were similar between the high-tension and low-tension groups, respectively.²⁰ Preoperatively, the AP laxity values between the 2 tension groups were not significantly different ($P = .70$), but they were significantly greater than in the control group ($P < .001$) (Figure 2A).²⁰

Clinical Outcomes

Immediately after graft fixation, mean AP laxity differences between limbs for the low-tension and high-tension groups were 0.2 mm (−0.1 to 0.5) and −2.1 mm (−2.3 to −1.9), respectively ($P < .001$), as intended by each initial graft tension protocol (Figure 2A). The AP laxity values between the 2 initial graft tension groups were not significantly different at 60-month ($P = .85$) or 84-month ($P = .83$) follow-up. At 84 months, there was a significant difference in laxity values between the high-tension and control groups ($P < .001$) but no significant difference between the low-tension and control groups ($P = .06$). As would be expected, there were no temporal changes in AP laxity in the control group ($P = .23$).

Preoperatively, the distributions of the IKDC examination scores between the 2 initial graft tension groups ranged from B (nearly normal) to D (severely abnormal), while scores in the control group ranged from A (normal) to B (nearly normal)

TABLE 1
International Knee Documentation Committee
Examination Scores for the 3 Study Groups^a

Score	Low-Tension Group	High-Tension Group	Control Group
Preoperatively			
A	0 (0.0)	0 (0.0)	58 (96.7)
B	27 (60.0)	27 (64.3)	2 (3.3)
C	17 (37.8)	13 (31.0)	0 (0.0)
D	1 (2.2)	2 (4.8)	0 (0.0)
36-month follow-up			
A	20 (60.6)	17 (53.1)	32 (86.5)
B	10 (30.3)	11 (34.4)	4 (10.8)
C	3 (9.1)	4 (12.5)	1 (2.7)
D	0 (0.0)	0 (0.0)	0 (0.0)
60-month follow-up			
A	13 (44.8)	10 (41.7)	21 (75.0)
B	13 (44.8)	12 (50.0)	7 (25.0)
C	3 (10.3)	2 (8.3)	0 (0.0)
D	0 (0.0)	0 (0.0)	0 (0.0)
84-month follow-up			
A	8 (25.0)	7 (24.1)	24 (75.0)
B	17 (53.1)	21 (72.4)	8 (25.0)
C	7 (21.9)	1 (3.5)	0 (0.0)
D	0 (0.0)	0 (0.0)	0 (0.0)

^aValues are expressed as n (%).

(Table 1). There was no significant difference in the distributions of the IKDC scores between the 2 initial graft tension groups ($P = .68$); however, they were significantly different from those in the control group ($P < .001$). Both the low-tension and high-tension groups remained significantly different from the control group at 60 months ($P = .04$ and $.03$, respectively) and 84 months ($P < .001$ for both) after surgery.

Functional Outcomes

Preoperatively, the hop distances were not significantly different between the 2 initial graft tension groups ($P =$

TABLE 2
Knee injury and Osteoarthritis Outcome Score Results^a

Subscale	Low-Tension Group	High-Tension Group	Control Group
Quality of life			
Preoperatively	34.5 (29.0-40.0) ^b	40.3 (34.6-46.0) ^b	98.5 (92.9-104.1)
60 months	80.5 (74.6-86.4) ^b	81.3 (75.3-87.3) ^b	94.6 (88.6-100.6)
84 months	76.5 (70.9-82.1) ^b	81.1 (76.8-86.8) ^b	92.3 (86.5-98.1)
Sports and recreation			
Preoperatively	55.4 (50.2-60.6) ^b	59.6 (54.2-65.0) ^b	98.0 (92.8-103.2)
60 months	85.3 (79.8-90.8) ^b	89.2 (83.6-94.8)	93.9 (88.3-99.5)
84 months	79.9 (74.7-85.1) ^{b,d}	87.8 (82.5-93.1) ^d	93.3 (87.9-98.7)
Activities of daily living			
Preoperatively	85.8 (83.2-88.4) ^b	86.0 (83.3-88.7) ^b	99.9 (97.2-102.6)
60 months	96.9 (94.1-99.7)	97.0 (94.1-99.9)	99.4 (96.5-102.3)
84 months	95.0 (92.3-97.7) ^c	96.6 (93.9-99.3)	98.7 (95.5-101.5)
Symptoms			
Preoperatively	69.6 (65.4-73.8) ^b	73.0 (68.7-77.3) ^b	96.3 (92.1-100.5)
60 months	85.3 (80.9-89.7) ^b	85.8 (81.3-90.3) ^b	93.0 (88.5-97.5)
84 months	82.5 (78.3-86.7) ^b	85.8 (81.5-90.1) ^b	92.4 (88.1-96.7)
Pain			
Preoperatively	75.1 (71.7-78.5) ^b	78.5 (75.0-82.0) ^b	99.0 (95.6-102.4)
60 months	91.4 (87.8-95.0) ^c	94.3 (90.6-98.0)	96.0 (92.3-99.7)
84 months	90.6 (87.2-94.0) ^b	92.9 (89.4-96.4)	96.3 (92.8-99.8)

^aValues are expressed as mean (95% CI). There was a statistically significant tension treatment effect ($P < .001$) and tension time effect ($P < .001$) for all subscales.

There were differences between the 2 initial graft tension groups and the control group at most time points (^b $P < .05$, ^c $P < .10$), and there was a statistically significant difference in the sports and recreation subscores between the tension groups at 84-month follow-up (^d $P < .05$).

.59), but both groups were significantly less than the control group ($P \leq .017$) (Figure 2B). After 84 months of healing, hop distances between the 2 initial graft tension groups were not significantly different from each other ($P = .80$). Both the low-tension ($P = .008$) and high-tension ($P = .017$) groups were significantly less than the control group. There were no temporal changes in hop distances in the control group ($P = .79$).

At 60 months, mean peak extension torques (normalized values) in the low-tension, high-tension, and control groups were 93.0% (89.6%-96.4%), 99.0% (95.4%-102.6%), and 94.9% (91.6%-98.2%), respectively; at 84 months, they were 99.7% (96.4%-104.0%), 102.4% (98.8%-106.0%), and 99.5% (96.3%-102.7%), respectively. There were no significant differences between the treatment and control groups at 60 months ($P = .22$) or 84 months ($P = .54$).

Patient-Reported Outcomes

The mean preinjury Tegner activity scores were 7.4 (6.8-8.0), 7.8 (7.2-8.4), and 6.4 (5.8-7.0) for the low-tension, high-tension, and control groups, respectively. At 60 months, these dropped to 5.4 (4.7-6.1), 6.2 (5.5-6.9), and 5.6 (4.9-6.3), respectively, and at 84 months to 4.8 (4.2-5.4), 5.6 (5.0-6.2), and 5.5 (4.9-6.1), respectively. There was a significant reduction in the mean Tegner scores over time in both tension groups as well as the control group ($P < .009$). There was no significant difference between the treatment groups at 60 months ($P = .12$) or 84 months ($P = .06$). Furthermore, the scores for the 2 initial graft tension groups were not significantly different

from those for the control group at any postoperative time point ($P > .12$).

Preoperatively, all KOOS values for the graft tension groups were significantly different from those for the control group ($P < .001$) (Table 2). At 84 months, all KOOS values in the low-tension group were significantly less than those for the control group ($P < .03$), with the exception of the KOOS-ADL subscale ($P = .062$). While the values for the high-tension group for the KOOS-QOL (Figure 2C) and KOOS-symptoms subscales were significantly less ($P < .03$) than those for the control group, there were no significant differences ($P > .15$) in the KOOS-sport, KOOS-ADL, or KOOS-pain subscales. Eight patients in the low-tension group, 5 in the high-tension group, and 1 in the control group had combined KOOS model outcomes indicative of symptomatic knee OA (low-tension vs control, $P = .01$; high-tension vs control, $P = .09$).^{18,44}

Two of the 8 SF-36 health domain scores (vitality and mental health) in the low-tension group were significantly lower than in the high-tension group ($P < .04$) at 60 months, although not at 84 months. The high-tension group scored better than the control group ($P = .02$) in 1 health domain (mental health) at 60 months, but there were no significant differences at 84 months. There were no significant differences between the low-tension and control groups at 60 months; however, for 2 of the health domains (general health and social functioning), the low-tension group scored significantly lower than the control group ($P < .04$) at 84 months (Table 3).

While there were no significant differences in the number of subsequent knee injuries between the treatment

TABLE 3
Short Form-36 Outcomes^a

Subscale	Low-Tension Group	High-Tension Group	Control Group
Physical functioning (tension treatment effect: $P < .001$; tension time effect: $P < .001$)			
Preoperatively	71.6 (69.6-73.6) ^b	73.3 (71.2-75.4) ^b	97.8 (95.8-99.8)
60 months	94.2 (92.1-96.3)	96.6 (94.4-98.8)	96.5 (94.4-98.6)
84 months	92.4 (90.4-94.4) ^c	95.1 (93.1-97.1)	97.8 (95.7-99.9)
Role physical (tension treatment effect: $P < .001$; tension time effect: $P < .001$)			
Preoperatively	61.7 (56.2-67.2) ^b	58.2 (55.3-61.1) ^b	97.0 (94.2-99.8)
60 months	97.5 (94.4-100.6)	96.7 (93.6-99.8)	97.2 (94.1-100.3)
84 months	94.8 (91.9-97.7)	97.6 (94.7-100.5)	96.6 (93.7-99.5)
Bodily pain (tension treatment effect: $P = .02$; tension time effect: $P < .001$)			
Preoperatively	63.5 (60.8-66.2) ^b	62.9 (60.1-65.7) ^b	84.8 (82.0-87.6)
60 months	82.5 (79.5-85.5)	88.3 (85.3-91.3)	82.3 (79.3-85.3)
84 months	78.2 (75.4-81.0) ^{c,e}	85.5 (82.7-88.3)	85.6 (82.7-88.5)
General health (tension treatment effect: $P = .22$; tension time effect: $P = .68$)			
Preoperatively	81.4 (79.3-83.5)	85.4 (83.3-87.5)	85.8 (83.7-87.9)
60 months	83.3 (81.1-85.5)	86.0 (83.7-88.3)	85.7 (83.5-87.9)
84 months	80.7 (78.6-82.8) ^b	84.1 (82.0-86.2)	86.9 (84.7-89.1)
Vitality (tension treatment effect: $P = .19$; tension time effect: $P = .02$)			
Preoperatively	62.3 (59.7-64.9) ^b	66.1 (63.4-68.8)	71.8 (69.1-74.5)
60 months	67.9 (65.1-70.7) ^d	76.3 (73.4-79.2) ^c	68.9 (66.1-71.7)
84 months	69.2 (66.5-71.9)	72.2 (69.5-74.9)	71.7 (68.9-74.5)
Social functioning (tension treatment effect: $P < .001$; tension time effect: $P < .001$)			
Preoperatively	82.2 (79.9-84.5) ^b	77.2 (74.8-79.6) ^b	97.0 (94.6-99.4)
60 months	92.4 (89.9-94.9)	94.6 (92.0-97.2)	97.3 (94.8-99.8)
84 months	90.3 (87.9-92.7) ^{b,e}	96.5 (94.1-98.9)	97.2 (94.7-99.7)
Emotional role (tension treatment effect: $P = .15$; tension time effect: $P < .001$)			
Preoperatively	83.8 (81.4-86.2) ^b	83.7 (81.2-86.2) ^b	97.1 (94.7-99.5)
60 months	93.3 (90.7-95.9)	96.0 (93.3-98.7)	96.7 (94.1-99.3)
84 months	95.5 (93.0-98.0)	95.6 (93.1-98.1)	97.2 (94.7-99.7)
Mental health (tension treatment effect: $P = .06$; tension time effect: $P < .001$)			
Preoperatively	73.4 (71.2-75.6) ^b	75.1 (72.9-77.3) ^b	83.9 (81.7-86.1)
60 months	78.4 (76.1-80.7) ^d	87.7 (85.3-90.1) ^b	80.0 (77.7-82.3)
84 months	80.5 (78.3-82.7)	85.0 (82.8-87.2)	85.0 (82.7-87.3)

^aValues are expressed as mean (95% CI).

Preoperatively, the low-tension and high-tension groups were significantly different from the control group in 7 and 6 of the 8 subscales, respectively (^b $P < .05$, ^c $P < .10$). There was a significant difference in the vitality and mental health subscores between the 2 tension groups at 60 months (^d $P < .05$, ^e $P < .10$).

groups at 84 months ($P = .53$), there was a significant difference between the high-tension and control groups ($P = .02$). In the ACL-reconstructed knees, there were 4 knee injuries (2 graft tears, 2 meniscal tears) in the low-tension group and 7 knee injuries (2 graft tears, 4 meniscal tears, 1 graft + meniscal tear) in the high-tension group. In the contralateral knee, there were 3 knee injuries (2 ACL tears, 1 meniscal tear) in the low-tension group and 3 knee injuries (3 ACL tears) in the high-tension group. There were 4 knee injuries (1 ACL tear, 1 medial collateral ligament tear, 1 meniscal tear, 1 patellar dislocation) in the matched control group.

In the patients who filled out the supplemental questionnaire, it was found that the low-tension and high-tension groups were very satisfied with their surgical outcome, with mean scores of 9.10 (7.4-10.8) and 9.30 (8.0-10.6), respectively ($P = .67$). Twenty of 21 patients in the low-tension group and 19 of 20 in the high-tension group indicated that they would elect to undergo ACL reconstruction surgery again if it were needed.

OA Imaging Outcomes

There were no significant differences in the mean minimum medial JSW of the injured knee between the treatment groups at 60 months ($P = .17$) or 84 months ($P = .08$). The medial JSW of the high-tension group was significantly greater than that of the control group at both 60 months ($P = .002$) and 84 months ($P = .003$); however, the JSW of the low-tension group was not significantly greater ($P = .07$ and $.22$, respectively) (Figure 3A).

The mean differences in the OARSI radiographic scores (injured knee – contralateral knee) between the tension groups were not significant at 60 months ($P = .29$) or 84 months ($P = .23$); however, there was a significant increase (ie, worse score) between the low-tension and control groups at both 60 months ($P = .03$) and 84 months ($P < .001$) and between the high-tension and control groups at 84 months ($P = .02$) (Figure 3B).

The mean WOMBS values for the injured knee were greater (ie, worse) than those of the control group both

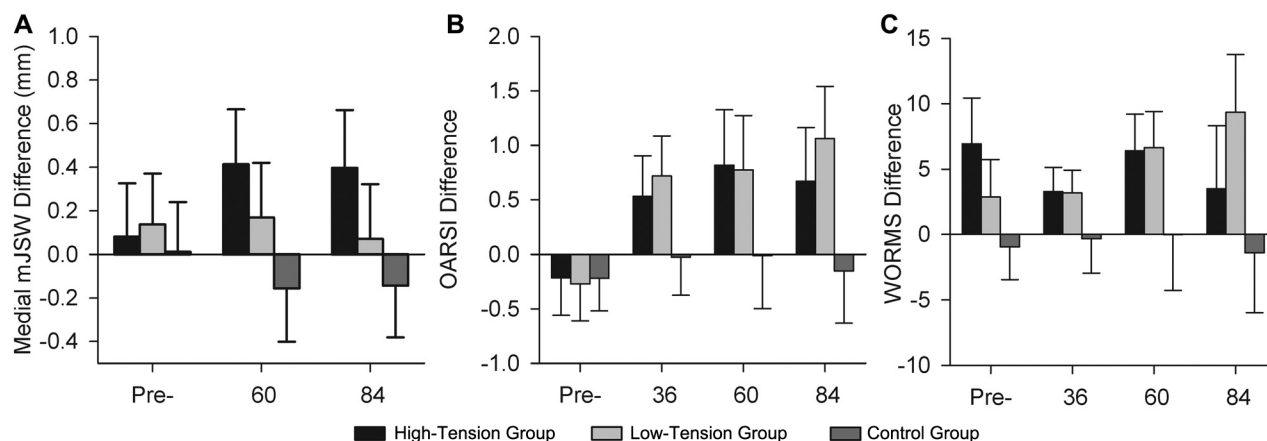


Figure 3. The mean differences between the operated and contralateral control knees for each group over 84 months with respect to the (A) minimum joint space width (mJSW) of the medial compartment, (B) Osteoarthritis Research Society International (OARS I) radiographic score, and (C) Whole-Organ Magnetic Resonance Imaging Score (WORMS). A positive value indicates that the operated knee is greater than the contralateral knee. Error bars represent 95% CI.

preoperatively and after 84 months (Figure 3C). The mean differences in the WORMS values (injured knee – contralateral knee) were not significantly different between the 2 initial graft tension treatment groups preoperatively ($P = .15$), at 60 months ($P = .69$), or at 84 months ($P = .08$). Both the high-tension and low-tension groups had significantly greater values than the control group at 60 months ($P = .03$ and $.01$, respectively). At 84 months, the WORMS values for the low-tension group remained significantly different compared with the control group ($P = .001$), but there was no significant difference between the high-tension and control groups ($P = .14$).

DISCUSSION

The most important findings of this study were that, in general, there was little difference in the clinical, functional, and patient-reported outcomes for the 2 initial graft tension groups. Furthermore, most outcomes of both groups were inferior to those of the control group at 84 months. Thus, the results do not support the hypotheses that the high-tension group would have better outcomes than the low-tension group and that the outcomes of the high-tension group would be equivalent to those of the control group. Nonetheless, the long-term follow-up has shown the emergence of differences in the KOOS-sport subscale and 2 of the SF-36 subscales between the 2 tension groups, with the high-tension group faring more than 8 points better than the low-tension group for all 3 scores—differences that were not present at 36 months.²⁰ An 8-point difference in the KOOS is thought to be clinically relevant.³⁸ The reason for the emerging differences is unclear but could possibly be the result of degenerative changes that are not detectable in the early stages of healing. Furthermore, the imaging outcomes evaluating OA suggest that knee arthrosis is progressing 84 months postoperatively in both treatment groups relative to the age-, sex-, race-, and activity-matched control group.

While several of the clinical and functional outcomes between the 2 tension groups were not significantly different, there were a few trends worth noting. First, there was a trend ($P = .089$) that the distribution of IKDC scores for the high-tension group was superior to that of the low-tension group (1 vs 7 C ratings, respectively) at 84 months. It is also interesting to note the trend that the mean Tegner score for the high-tension group was greater than that of the low-tension group at 84 months ($P = .06$). The activity level of the high-tension group was equivalent to that of the control group, while the low-tension group was less active. For the low-tension group, all of the KOOS values except for the KOOS-ADL subscale were significantly less than those of the control group, while for the high-tension group, only the KOOS-symptoms and KOOS-QOL values were less than those of the control group. It should be noted that 8 of the patients in the low-tension group and 5 in the high-tension group had KOOS values indicative of symptomatic knee OA according to the combined KOOS model of Englund et al¹⁸ in which the low-tension group reached statistical significance. While many of these outcomes represent trends, these were not present at the 36-month follow-up,²⁰ which further suggests that differences are emerging. Long-term follow-up will be required to determine the implications of these trends.

The imaging outcomes for OA suggest that, on average, knee arthrosis is progressing 84 months postoperatively. The minimum JSW of the medial compartment was measured with the validated method used by the Osteoarthritis Initiative.^{14,15} While not significant, there was a trend ($P = .08$) indicating that the minimum JSW of the surgical knee of the high-tension group was greater than that of the low-tension group at 84 months. These findings suggest that cartilage thickness was increasing in the high-tension group but not changing in the low-tension group. While advanced OA is typically marked by a decrease in articular cartilage thickness, an increase in cartilage thickness has been reported in the early stages of OA progression and

is thought to be caused by the breakdown of proteoglycan, which attracts water into the articular cartilage.^{10,36} A mean increase in articular cartilage thickness in ACL-reconstructed knees relative to the uninjured contralateral control knee has been reported by Jones et al²⁶ at 2 years after surgery. They also found that ACL-reconstructed knees that also underwent partial meniscectomy had narrower JSW measurements relative to the contralateral uninjured knee than knees of patients that underwent isolated ACL reconstruction, suggesting that knees with concomitant meniscal damage are further along the disease process.²⁶ It may be that the low-tension group is normal or that we are seeing OA development at different stages in both groups (ie, high-tension group at an earlier stage of OA progression). Additional longitudinal study is warranted.

While minimum JSW is a surrogate for articular cartilage thickness, the secondary cartilage imaging outcomes (OARSI and WOMS results) reflect overall joint health as it relates to OA because it is a disease that affects the whole joint. The OARSI radiographic scores were primarily influenced by osteophyte formation at 84 months. The WOMS values were affected by cartilage signal and osteophyte formation at 84 months, whereas at the time of injury (preoperative assessment), they were being driven by bone marrow edema, meniscus status, and ACL integrity. Longer term follow-up is required to determine if joint arthrosis will continue to progress in this cohort of ACL-reconstructed patients with isolated ACL injuries and minimal meniscal damage at the time of surgery. It is important to focus the study on patients with isolated ACL injuries as concomitant meniscal injuries have been shown to accelerate posttraumatic OA in ACL-injured patients.²⁶

There have been several randomized trials evaluating the effects of initial graft tension on clinical outcomes after single-bundle ACL reconstruction. For hamstring tendon grafts, Yasuda et al⁴⁸ reported that mean side-to-side differences in anterior laxity were significantly less when the graft was tensioned to 80 N compared with 20 N at ≥ 2 years after surgery. However, Kim et al²⁸ found no significant differences in knee laxity when the grafts were tensioned to 78, 117, and 146 N after 1 year. For patellar tendon grafts, Nicholas et al³² reported significant improvement in knee laxity when grafts were tensioned to 90 N as compared with 45 N. However, Yoshiya et al⁴⁹ and van Kampen et al⁴¹ found no differences in knee laxity. While 2 of these studies showed differences in knee laxity, no differences in any other outcomes were found. In an evidence-based review, it was concluded that “there is no clear trend in terms of statistically significant or clinically relevant differences in the amount of tension to apply to the graft during graft fixation.”⁴ Likewise, in the present study, there were no significant differences in most of the outcomes between the 2 laxity-based initial graft tension protocols through 84 months. The comparison with the control group, which was unique to the current study, also indicated that many of the outcome measures were inferior to those of the matched control group despite the initial graft tension level.

A laxity-based initial graft tensioning protocol was selected for study instead of a “force-based” approach.¹⁹ The laxity-based approach implies that the tension in the

graft is adjusted to produce an AP laxity value that is equal to or less than that of the contralateral normal knee. A force-based approach (eg, 20 N, 40 N) was not used for this study because the amount of force required to restore AP knee laxity is patient specific⁹ and prone to error.^{21,42} Although a laxity-based tensioning approach may require a variety of tension levels across patients within a treatment group, the tension applied to the graft in an effort to re-create normal AP knee laxity would be less than that required to overconstrain the joint within the same patient, hence the terminology of “low tension” versus “high tension.”

For this trial, the grafts were tensioned to match normal AP laxity or overconstrain it by 2 mm. The desired initial conditions were achieved; however, both grafts stretched out over time. After 7 years of healing, the mean difference between the surgical limb and the contralateral control limb across the 2 treatment groups was approximately 1.5 mm, while the knees of the control group did not stretch out over time (Figure 2A). Most of the increase occurred over the first 6 months of healing. The amount of increase observed was similar to those reported for other clinical studies of ACL reconstruction.^{5,29} Although the results suggest that the differences in AP knee laxity between the 2 laxity-based initial graft tension conditions are minimal at 84 months, differences in joint contact conditions immediately after surgery may still affect long-term joint health. Recent evidence suggests that the inflammatory response after an ACL injury may have a negative effect on articular cartilage,^{22,30} which in turn may compromise a primary lubricating mechanism of the joint.^{16,17} Although details about the interactions between inflammatory cytokines, degradative enzymes, lubrication, and OA remain to be worked out, cartilage is thought to be at high risk for damage in the early days after an injury.^{3,11,22} Surgery also adds a second inflammatory insult, which could exacerbate the problem.²³ High contact stresses and increased shear motion during early healing, when catabolic enzymes are present in synovial fluid, could promote OA.³ Thus, it is likely that initial differences in joint contact mechanics would influence long-term joint health. At this time, it appears that OA may be progressing in both initial graft tension groups compared with the uninjured control group.

There are several limitations to this study. As with any long-term follow-up, the number of participants who drop out will introduce bias. However, the number of patients who dropped out of the tension cohorts at 84 months was only 20%. Loss to follow-up would also affect the power of the study; however, the change was minimal as the study had power to detect an 18% difference in knee laxity, a 13.5% change in the IKDC score, a 13.5% change in peak torque, and a 12% change in the KOOS value with the reduced number. The dropout rate for the control group was higher; however, we overrecruited the control group by 15 participants to compensate for the expected higher loss to follow-up. While the control group was relatively small ($n = 35$ at 84 months), matching by age, sex, race, and activity was performed to try to construct a representative control population for comparison. It should also be

noted that the supplemental patient satisfaction questionnaire was completed by only 41 of the 72 eligible participants, as it was mailed out after all participants had returned their original questionnaires.

Structural assessments of the JSW via radiography are not very sensitive measures to detect changes in the early stages of OA.³⁷ While joint space narrowing in the medial compartment did not occur, on average, in our cohort at 84 months, the OARSI radiographic score² and WOMS³⁵ are outcomes that focus on the whole joint and not just cartilage alone. These latter measures are starting to show differences between the experimental groups. It is likely that joint space narrowing will follow given enough time. While the radiologist was blinded to the treatment group, it is possible that the presence of screws and tunnels in the surgical knee could bias scoring of the radiographs or magnetic resonance imaging scans. However, he would be unable to distinguish images between the contralateral knees of the treatment group and the knees of the control group or between reconstructed knees of the 2 graft tension groups.

In this study, our objective was to establish the progression of OA in the ACL-reconstructed knee after an isolated ACL injury while accounting for confounding variables including graft type, graft placement, concomitant injuries, and postoperative rehabilitation. Therefore, only patients receiving autografts were enrolled, all surgeons used the same drill guide system, patients with concomitant injuries (other than a minor meniscal tear) were excluded, and all patients were prescribed the same postoperative rehabilitation program. A limitation is that we included both patellar tendon and 4-stranded hamstring tendon autografts in the study. Because of the popularity of these 2 autografts, the participating surgeons felt that it was important to offer the patient his or her choice. This was not seen as a significant limitation as previous biomechanical and clinical studies found them to be equivalent.^{31,45,46} However, a recent meta-analysis suggests that the radiographic incidence of OA after ACL reconstruction is greater for bone-patellar tendon-bone autografts compared with hamstring tendon autografts.⁴⁷ As graft type was not randomized in this study, and because only a third of the patients received a hamstring tendon graft, the study was not powered to address the question. Nonetheless, the ratio of graft types in both treatment groups was similar, which would minimize bias.²⁰

In conclusion, the results do not support our hypotheses that the high-tension group would have improved outcomes when compared with the low-tension group after 84 months of healing and that the outcomes for the high-tension group would be equivalent to those for the matched control group. While there were minor differences in patient-reported outcomes between the 2 laxity-based tension groups, all other outcomes were similar. Long-term data are needed to reconcile these findings.

ACKNOWLEDGMENT

The authors gratefully acknowledge the assistance of Melanie Westwell, Nabil Mehta, Alison Biercevicz, Emily

Robbins, Heidi Oksendahl, Daniel Acuri, Lynn Fanella, and David Dimascio. The magnetic resonance imaging (MRI) scans were acquired at the MRI Research Facility (Brown University, Providence, Rhode Island) and the Osteoarthritis Initiative MRI Facility (Southeastern Medical Center, Memorial Hospital of Rhode Island, Pawtucket, Rhode Island).

REFERENCES

1. Ajuied A, Wong F, Smith C, et al. Anterior cruciate ligament injury and radiologic progression of knee osteoarthritis: a systematic review and meta-analysis. *Am J Sports Med*. 2014;42:2242-2252.
2. Altman RD, Gold GE. Atlas of individual radiographic features in osteoarthritis, revised. *Osteoarthritis Cartilage*. 2007;15:A1-A56.
3. Anderson DD, Chubinskaya S, Guilak F, et al. Post-traumatic osteoarthritis: improved understanding and opportunities for early intervention. *J Orthop Res*. 2011;29:802-809.
4. Arneja S, McConkey MO, Mulpuri K, et al. Graft tensioning in anterior cruciate ligament reconstruction: a systematic review of randomized controlled trials. *Arthroscopy*. 2009;25:200-207.
5. Beynnon BD, Johnson RJ, Naud S, et al. Accelerated versus nonaccelerated rehabilitation after anterior cruciate ligament reconstruction: a prospective, randomized, double-blind investigation evaluating knee joint laxity using roentgen stereophotogrammetric analysis. *Am J Sports Med*. 2011;39:2536-2548.
6. Beynnon BD, Uh BS, Fleming BC, et al. Rehabilitation following anterior cruciate ligament reconstruction: a prospective, randomized, double-blind comparison of accelerated versus non-accelerated rehabilitation. *Am J Sports Med*. 2005;33:347-359.
7. Biercevicz AM, Akelman MR, Fadale PD, et al. MRI volume and signal intensity of ACL graft predict clinical, functional, and patient-oriented outcome measures after ACL reconstruction. *Am J Sports Med*. 2015;43:693-699.
8. Buckland-Wright JC, Wolfe F, Ward RJ. Substantial superiority of semiflexed (MTP) views in knee osteoarthritis: a comparative radiographic study, without fluoroscopy, of standing extended, semiflexed (MTP), and Schuss views. *J Rheumatol*. 1999;26:2664-2674.
9. Burks RT, Leland R. Determination of graft tension before fixation in anterior cruciate ligament reconstruction. *Arthroscopy*. 1988;4:260-266.
10. Calvo E, Palacios I, Delgado E, et al. Histopathological correlation of cartilage swelling detected by magnetic resonance imaging in early experimental osteoarthritis. *Osteoarthritis Cartilage*. 2004;12:878-886.
11. Catterall JB, Stabler TV, Flannery CR, Kraus VB. Changes in serum and synovial fluid biomarkers after acute injury (NCT00332254). *Arthritis Res Ther*. 2010;12:R229.
12. Daniel DM, Malcom LL, Losse GM, Stone ML, Sachs R, Burks RT. Instrumented measurement of anterior laxity of the knee. *J Bone Joint Surg Am*. 1985;67:720-726.
13. Duryea J, Li J, Peterfy CG, Gordon C, Genant HK. Trainable rule-based algorithm for the measurement of joint space width in digital radiographic images of the knee. *Med Phys*. 2000;27:580-591.
14. Duryea J, Neumann G, Niu J, et al. Comparison of radiographic joint space width with magnetic resonance imaging cartilage morphometry: analysis of longitudinal data from the Osteoarthritis Initiative. *Arthritis Care Res (Hoboken)*. 2010;62:932-937.
15. Duryea J, Zaim S, Genant HK. New radiographic-based surrogate outcome measures for osteoarthritis of the knee. *Osteoarthritis Cartilage*. 2003;11:102-110.
16. Elsaid KA, Jay GD, Chichester CO. Reduced expression and proteolytic susceptibility of lubricin/superficial zone protein may explain early elevation in the coefficient of friction in the joints of rats with antigen-induced arthritis. *Arthritis Rheum*. 2007;56:108-116.
17. Elsaid KA, Machan JT, Waller K, Fleming BC, Jay GD. The impact of anterior cruciate ligament injury on lubricin metabolism and the effect of inhibiting tumor necrosis factor alpha on chondroprotection in an animal model. *Arthritis Rheum*. 2009;60:2997-3006.

18. Englund M, Roos EM, Lohmander LS. Impact of type of meniscal tear on radiographic and symptomatic knee osteoarthritis: a sixteen-year followup of meniscectomy with matched controls. *Arthritis Rheum.* 2003;48:2178-2187.
19. Fleming BC, Brady MF, Bradley MP, Banerjee R, Hulstyn MJ, Fadale PD. Tibiofemoral compression force differences using laxity- and force-based initial graft tensioning techniques in the anterior cruciate ligament-reconstructed cadaveric knee. *Arthroscopy.* 2008;24:1052-1060.
20. Fleming BC, Fadale PD, Hulstyn MJ, et al. The effect of initial graft tension after anterior cruciate ligament reconstruction: a randomized clinical trial with 36-month follow-up. *Am J Sports Med.* 2013;41:25-34.
21. Fleming BC, Peura GD, Beynon BD. Factors influencing the output of an implantable force transducer. *J Biomech.* 2000;33:889-893.
22. Haslauer CM, Elsaid KA, Fleming BC, Proffen BL, Johnson VM, Murray MM. Loss of extracellular matrix from articular cartilage is mediated by the synovium and ligament after anterior cruciate ligament injury. *Osteoarthritis Cartilage.* 2013;21:1950-1957.
23. Heard BJ, Solbak NM, Achari Y, et al. Changes of early post-traumatic osteoarthritis in an ovine model of simulated ACL reconstruction are associated with transient acute post-injury synovial inflammation and tissue catabolism. *Osteoarthritis Cartilage.* 2013;21:1942-1949.
24. Hunter DJ, Arden N, Conaghan PG, et al. Definition of osteoarthritis on MRI: results of a Delphi exercise. *Osteoarthritis Cartilage.* 2011;19:963-969.
25. Irrgang JJ, Ho H, Harner CD, Fu FH. Use of the International Knee Documentation Committee guidelines to assess outcome following anterior cruciate ligament reconstruction. *Knee Surg Sports Traumatol Arthrosc.* 1998;6:107-114.
26. Jones MH, Spindler KP, Fleming BC, et al. Meniscus treatment and age associated with narrower radiographic joint space width 2 - 3 years after ACL reconstruction: data from the MOON onsite cohort. *Osteoarthritis Cartilage.* 2015;23:581-588.
27. Jones MH, Spindler KP, Fleming BC, et al. Predictors of lateral compartment joint space difference two or more years after ACL reconstruction: data from the MOON onsite cohort. *Osteoarthritis Cartilage.* 2015;23:A280.
28. Kim SG, Kurosawa H, Sakuraba K, Ikeda H, Takazawa S. The effect of initial graft tension on postoperative clinical outcome in anterior cruciate ligament reconstruction with semitendinosus tendon. *Arch Orthop Trauma Surg.* 2006;126:260-264.
29. Magnussen RA, Carey JL, Spindler KP. Does autograft choice determine intermediate-term outcome of ACL reconstruction? *Knee Surg Sports Traumatol Arthrosc.* 2011;19:462-472.
30. McNulty AL, Rothfusz NE, Leddy HA, Guilak F. Synovial fluid concentrations and relative potency of interleukin-1 alpha and beta in cartilage and meniscus degradation. *J Orthop Res.* 2013;31:1039-1045.
31. Mohtadi NG, Chan DS, Dainty KN, Whelan DB. Patellar tendon versus hamstring tendon autograft for anterior cruciate ligament rupture in adults. *Cochrane Database Syst Rev.* 2011;(9):CD005960.
32. Nicholas SJ, D'Amato MJ, Mullaney MJ, Tyler TF, Kolstad K, McHugh MP. A prospectively randomized double-blind study on the effect of initial graft tension on knee stability after anterior cruciate ligament reconstruction. *Am J Sports Med.* 2004;32:1881-1886.
33. Patel RR, Hurwitz DE, Bush-Joseph CA, Bach BR, Andriacchi TP. Comparison of clinical and dynamic knee function in patients with anterior cruciate ligament deficiency. *Am J Sports Med.* 2003;31:68-74.
34. Peterfy C, Li J, Zaim S, et al. Comparison of fixed-flexion positioning with fluoroscopic semi-flexed positioning for quantifying radiographic joint-space width in the knee: test-retest reproducibility. *Skeletal Radiology.* 2003;32:128-132.
35. Peterfy CG, Guermazi A, Zaim S, et al. Whole-Organ Magnetic Resonance Imaging Score (WORMS) of the knee in osteoarthritis. *Osteoarthritis Cartilage.* 2004;12:177-190.
36. Qazi AA, Folkesson J, Pettersen PC, Karsdal MA, Christiansen C, Dam EB. Separation of healthy and early osteoarthritis by automatic quantification of cartilage homogeneity. *Osteoarthritis Cartilage.* 2007;15:1199-1206.
37. Reinke EK, Spindler KP, Lorrington D, et al. Hop tests correlate with IKDC and KOOS at minimum of 2 years after primary ACL reconstruction. *Knee Surg Sports Traumatol Arthrosc.* 2011;19:1806-1816.
38. Roos EM, Lohmander LS. The Knee injury and Osteoarthritis Outcome Score (KOOS): from joint injury to osteoarthritis. *Health Qual Life Outcomes.* 2003;1:64.
39. Roos EM, Roos HP, Lohmander LS, Ekdahl C, Beynon BD. Knee Injury and Osteoarthritis Outcome Score (KOOS): development of a self-administered outcome measure. *J Orthop Sports Phys Ther.* 1998;28:88-96.
40. Tegner Y, Lysholm J. Rating systems in the evaluation of knee ligament injuries. *Clin Orthop Relat Res.* 1985;198:43-49.
41. van Kampen A, Wymenga AB, van der Heide HJL. The effect of different graft tensioning in anterior cruciate ligament reconstruction: a prospective randomized study. *Arthroscopy.* 1998;14:845-850.
42. Ventura CP, Wolchok J, Hull ML, Howell SM. An implantable transducer for measuring tension in an anterior cruciate ligament graft. *J Biomech Eng.* 1998;120:327-333.
43. Ware JE, Sherbourne CD. The MOS 36-item Short-Form Health Survey (SF-36), I: conceptual framework and item selection. *Med Care.* 1992;30:473-483.
44. Wasserstein D, Huston LJ, Nwosu S, Group M, Spindler KP. KOOS pain as a marker for significant knee pain two and six years after primary ACL reconstruction: a Multicenter Orthopaedic Outcomes Network (MOON) prospective longitudinal cohort study. *Osteoarthritis Cartilage.* 2015;23:1674-1684.
45. Wilson TW, Zafuta MP, Zobitz M. A biomechanical analysis of matched bone-patellar tendon-bone and double-looped semitendinosus and gracilis tendon grafts. *Am J Sports Med.* 1999;27:202-207.
46. Woo SL-Y, Kanamori A, Zeminski J, Yagi M, Papageorgiou CD, Fu FH. The effectiveness of reconstruction of the anterior cruciate ligament with hamstrings and patellar tendon: a cadaveric study comparing anterior tibial and rotational loads. *J Bone Joint Surg Am.* 2002;84:907-914.
47. Xie X, Xiao Z, Li Q, et al. Increased incidence of osteoarthritis of knee joint after ACL reconstruction with bone-patellar tendon-bone autografts than hamstring autografts: a meta-analysis of 1,443 patients at a minimum of 5 years. *Eur J Orthop Surg Traumatol.* 2015;25:149-159.
48. Yasuda K, Tsujino J, Tanabe Y, Kaneda K. Effects of initial graft tension on clinical outcome after anterior cruciate ligament reconstruction: autogenous doubled hamstring tendons connected in series with polyester tapes. *Am J Sports Med.* 1997;25:99-106.
49. Yoshiya S, Kurosaka M, Ouchi K, Kuroda R, Mizuno K. Graft tension and knee stability after anterior cruciate ligament reconstruction. *Clin Orthop Relat Res.* 2002;394:154-160.

For reprints and permission queries, please visit SAGE's Web site at <http://www.sagepub.com/journalsPermissions.nav>.